

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Which logic gate outputs HIGH only when all inputs are HIGH?		1
A) OR		
B) AND		
C) NAND		
D) NOR		
2 Which code is used for error detection and correction?		1
A) Gray code		
B) Excess-3 code		
C) Hamming code		
D) ASCII		
3 K-map is used to:		1
A) Build microprocessors		
B) Generate flip-flops		
C) Simplify Boolean expressions		
D) Store digital data		
4 A 4-to-16 decoder has how many inputs and outputs?		1
A) 4 inputs, 16 outputs		
B) 16 inputs, 4 outputs		
C) 2 inputs, 8 outputs		
D) 8 inputs, 2 outputs		
5 The basic memory element in sequential logic is:		1
A) Multiplexer		
B) Decoder		
C) Flip-Flop		
D) Encoder		
6 JK flip-flop avoids race condition by:		1
A) Using only one input		
B) Feeding output back to input		
C) Using a clock pulse		
D) Avoiding logic gates		

- 7 What is the main function of a shift register? 1
 A) Arithmetic calculation
 B) Storage
 C) Data conversion
 D) Data shifting
- 8 The ALU in a microprocessor performs: 1
 A) Only addition
 B) Arithmetic and logical operations
 C) Memory access
 D) Clock generation
- 9 How many address lines are used by the 8086 microprocessor? 1
 A) 8
 B) 16
 C) 20
 D) 32
- 10 DMA stands for: 1
 A) Direct Machine Access
 B) Direct Memory Access
 C) Digital Memory Address
 D) Device Management Architecture
- 11 Which addressing mode uses a constant as an operand? 1
 A) Direct
 B) Indirect
 C) Immediate
 D) Indexed
- 12 The function of the Program Counter (PC) is to: 1
 A) Store data
 B) Store result of ALU
 C) Hold the address of next instruction
 D) Point to the stack

Q. 2 Solve the following. 12

- A)** Convert the following (L2) 6
 a) $(1AD)_{16} = ()_{10}$
 b) $(453)_8 = ()_{10}$
 c) $(10110011)_2 = ()_{10}$
- B)** Design a 4:1 multiplexer and explain its working with a logic diagram and truth table. (L3) 6

Q.3 Solve the following. 12

- A)** Design the circuit by Using NAND gates $F = ABC' + DE + AB'D'$ (L3) 6
- B)** Perform the following operations using 2's complement method (L3) 6
 (i) $(1101)_2 + (1001)_2$,
 (ii) $(1010)_2 - (1101)_2$

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Q. 4 Solve Any Two of the following.

- A) Design a 3 bit odd parity generator and implement with NAND gates. (L6) 6
- B) Differentiate between Synchronous & Asynchronous counters. Draw and explain 4-bit Ripple counter with waveforms (L4) 6
- C) Simplify the Boolean expression using K-MAP (L4) 6
 $F(A,B,C,D) = \sum m (1,2,3,8,9,10,11,14) + d (7,15)$

Q.5 Solve Any Two of the following.

12

- A) Draw the architecture of 8086 and explain the function of ALU, Bus Interface Unit, and Execution Unit. (L2) 6
- B) Design a 3-bit asynchronous counter using JK flip-flops. Explain its timing diagram. (L3) 6
- C) Explain the working of a Serial-In Parallel-Out shift register. (L2) 6

Q. 6 Solve Any Two of the following.

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- A) Write an 8086 assembly language program to add two 8-bit numbers stored at memory locations 3000H and 3001H. Store the result in 3002H. (L3) 6
- B) Explain the different addressing modes in 8086 with suitable examples. (L2) 6
- C) Draw and explain the working of a clocked SR flip-flop. Compare it with D flip-flop. (L4) 6

*** End ***